Mobile games analytics case study: UA, monetization, LTV

OVERVIEW

Five questions a UA, Ads & Portfolio analyst answers every week, worked end to end with SQL (DuckDB), Python (A/B test, LTV forecasting) and Power BI.

PUBLIC · PAGE 1

32%

of Rovio's gross bookings went to user acquisition in 2022, up from 23% in 2020

SIMULATED · PAGE 2

Day 306

AppLovin pays back its CPI; TikTok never does at its current CPI

SIMULATED · PAGE 4

< 6%

error on day-90 LTV when forecasting from the first 30 days only

REAL · PAGE 5

-0.8 pts

D7 retention if the first gate moves from level 30 to 40 (p = 0.002)

Business questions

1 · Portfolio How did Rovio grow bookings 2020-2022, and at what UA cost?
2 · UA Which acquisition channels pay back, and when?
3 · Retention How do cohorts retain, and where does revenue come from?
4 · LTV Can a channel be judged after 30 days instead of 90+?
5 · A/B test Should the first gate sit at level 30 or 40?

Recommendation: shift UA budget toward AppLovin and Google, cap TikTok until its CPI drops below ~€0.93 (its 1-year LTV), review new channels at day 30, and keep the gate at level 30.

Data & methods

PUBLIC Rovio quarterly KPIs and bookings by game, from published reports.
SIMULATED Players, activity and UA spend generated in Python. Realistic patterns, not Rovio's numbers.
REAL Cookie Cats A/B test, 90,189 players.

SQL censoring-safe cohort retention, cumulative ARPI, ROAS by channel (DuckDB).
Python z-test + bootstrap; power-curve LTV fit a·t^b with D30→D90 holdout; payback day.
Power BI star schema, 30 DAX measures, one question per page.

How to read this report

Titles state the conclusion of the page. **Tags** PUBLIC / SIMULATED / REAL mark where the data comes from. **Dashed lines** mark break-even or a threshold. **Code** SQL, Python notebook and Power BI project in the repository.

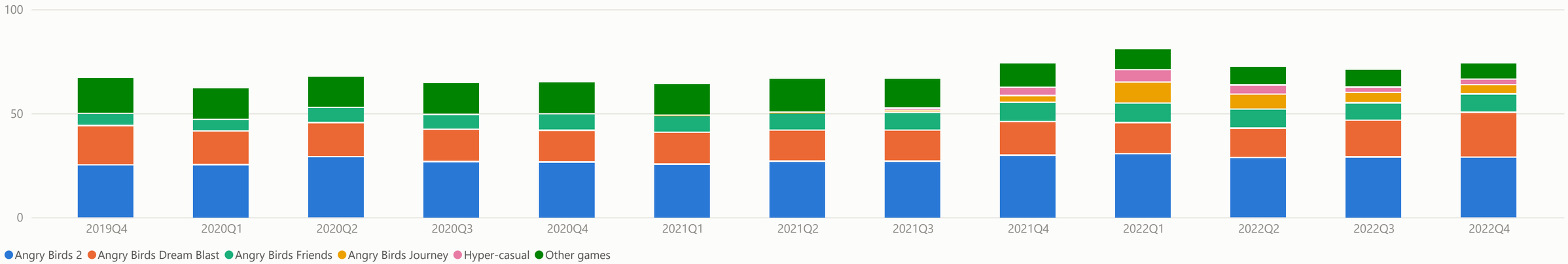
Rovio portfolio: growth was bought, then monetized better

PUBLIC DATA

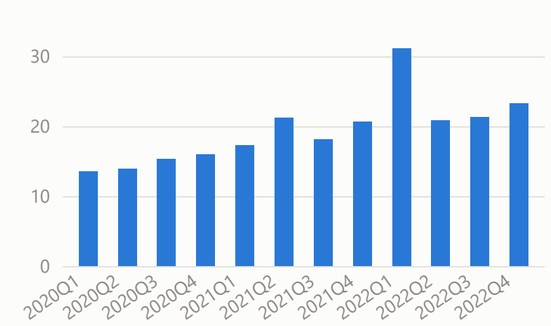
UA rose from 23% of gross bookings in 2020 to 32% in 2022, while top-5 ARPDAU rose from €0.18 to €0.21 (annual figures). Dream Blast's Q4-22 record quarter came with the heaviest UA push.

Gross bookings 2022 (EUR m)	User acquisition 2022 (EUR m)	UA share of gross bookings, 2022	ARPDAU top-5 games, 2022 average
298,0	96,5	32,4%	€0,21

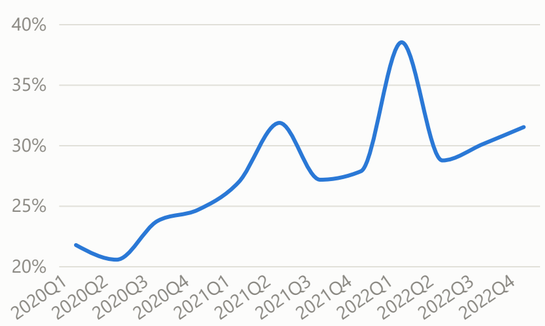
Gross bookings by game, EUR m (stacked)



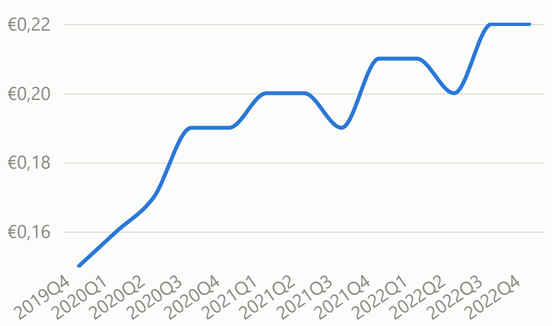
UA spend, EUR m



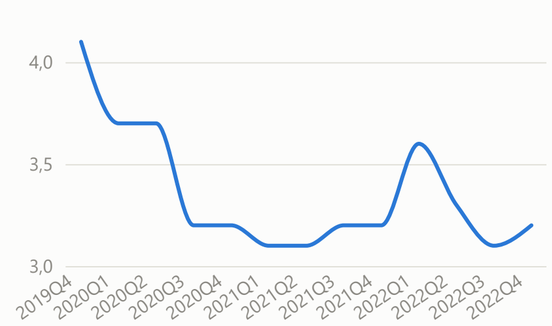
UA as % of gross bookings



ARPDAU top-5 games, EUR



DAU top-5 games, millions



Source: Rovio Entertainment interim reports & financial statements 2021-2022 (public) · Independent Data Analyst case study · Not affiliated with Rovio

AppLovin costs the most per install and pays back best; TikTok never pays back

SIMULATED DATA

ROAS compares revenue generated by a channel's installs with what was spent to buy them. Above 100% the channel has paid for itself. AppLovin breaks even around day 306.

Total UA spend

€139K

Paid installs

51K

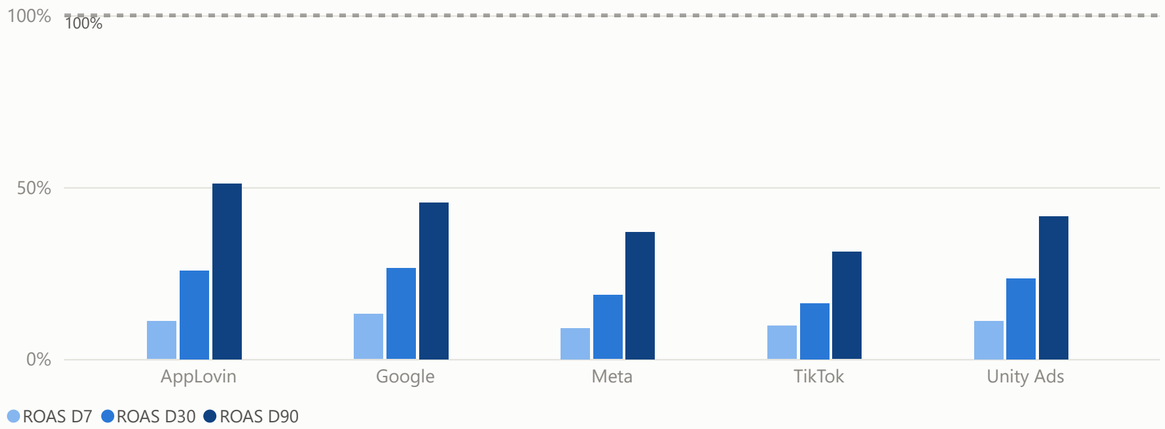
Blended CPI

€2,74

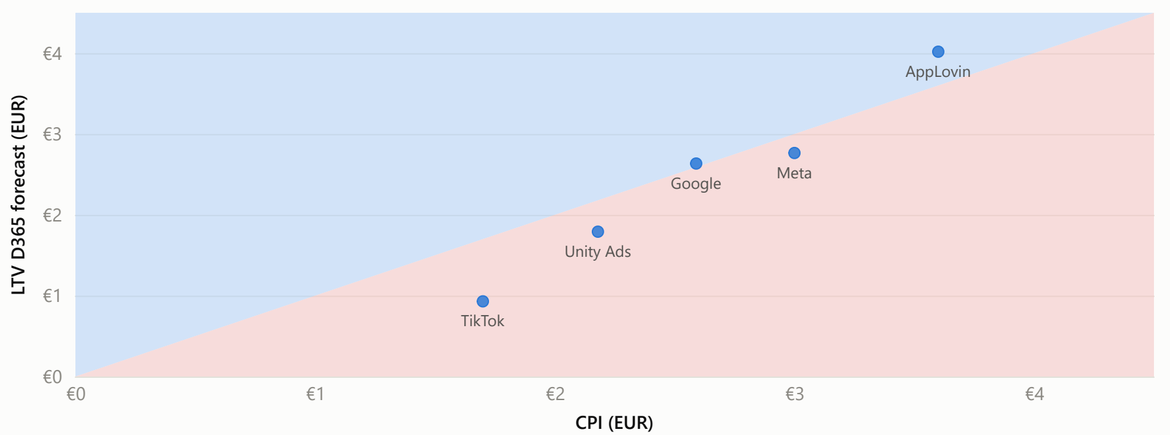
ROAS D90 (spend-weighted)

43%

ROAS by channel at D7 / D30 / D90 (dashed = break-even)



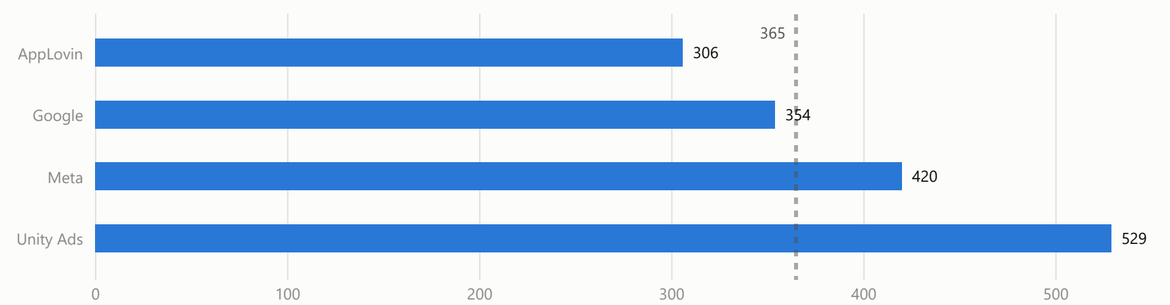
CPI vs forecast 1-year LTV per install, EUR (above the diagonal: LTV > CPI, pays back within a year)



Channel scorecard (red = below break-even, blue = above)

Channel	CPI	ROAS D90	ROAS D365 Forecast	Payback day (forecast)
AppLovin	€3,60	51%	112%	306
Google	€2,58	46%	102%	354
Meta	€2,99	37%	92%	420
TikTok	€1,70	31%	55%	
Unity Ads	€2,20	42%	82%	529

Forecast payback day by channel (dashed = 1 year; TikTok never pays back)



Simulated player-level data (seed 42), illustrative only - not Rovio data · Independent Data Analyst case study · Not affiliated with Rovio

Retention falls from 47% at D1 to 21% at D7, then flattens; ads bring 23% of revenue

SIMULATED DATA

Most churn happens in the first week (D30: 11%, D90: 7%), so early-game and onboarding changes have the biggest lever. Ads are a meaningful second revenue stream next to in-app purchases.

Channel

Tout

Country

Tout

Platform

Tout

IAP revenue

€100K

Ad revenue

€29K

Ad share of revenue

23%

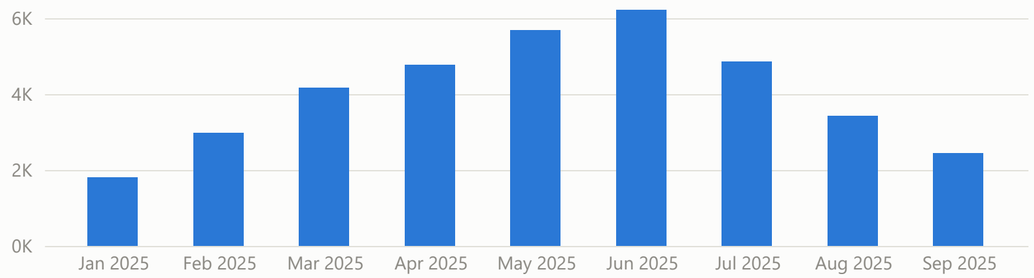
Payer conversion

1,7%

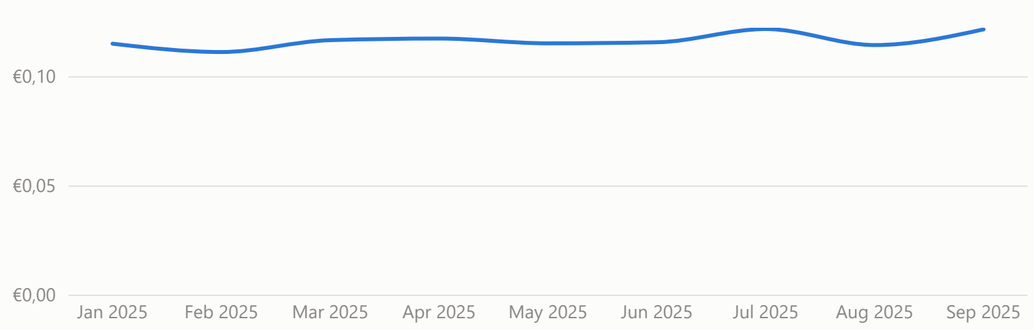
Cohort retention by install week and day N (all players, not filtered by slicers)

Install week	1	3	7	14	30	60	90
30 déc. 2024	46,1%	29,9%	20,4%	15,4%	10,8%	7,9%	6,1%
06 janv. 2025	46,9%	28,9%	20,0%	14,3%	10,5%	7,5%	6,5%
13 janv. 2025	46,0%	28,5%	20,2%	14,9%	10,6%	8,3%	7,2%
20 janv. 2025	46,1%	29,1%	21,1%	15,6%	11,3%	7,9%	7,2%
27 janv. 2025	47,2%	29,1%	20,4%	14,1%	10,4%	7,4%	6,6%
03 févr. 2025	46,4%	30,0%	21,6%	15,6%	10,4%	8,6%	6,8%
10 févr. 2025	46,6%	30,7%	21,3%	15,5%	9,6%	7,5%	6,6%
17 févr. 2025	47,2%	28,9%	20,8%	16,0%	11,3%	9,2%	7,7%
24 févr. 2025	46,3%	28,3%	20,4%	15,7%	10,6%	8,6%	6,9%
03 mars 2025	47,4%	29,4%	21,1%	15,2%	10,7%	8,4%	6,5%
10 mars 2025	47,9%	29,7%	19,9%	14,7%	10,2%	8,1%	6,5%
17 mars 2025	47,6%	31,2%	22,6%	16,4%	11,0%	7,9%	7,0%
24 mars 2025	46,2%	29,5%	20,8%	15,0%	10,8%	7,4%	6,5%
31 mars 2025	46,2%	30,3%	21,7%	15,2%	11,1%	8,6%	7,3%
07 avr. 2025	46,2%	30,7%	19,4%	13,6%	9,8%	7,9%	6,3%
14 avr. 2025	46,3%	30,5%	19,7%	15,4%	11,6%	8,5%	7,0%
21 avr. 2025	46,3%	30,7%	21,6%	15,8%	12,0%	8,5%	7,0%
28 avr. 2025	45,0%	28,8%	20,2%	14,3%	11,0%	8,1%	6,3%

Average DAU by month (simulated installs stop on 30 June, so DAU declines from July)



ARPDAU by month, EUR (axis from 0)

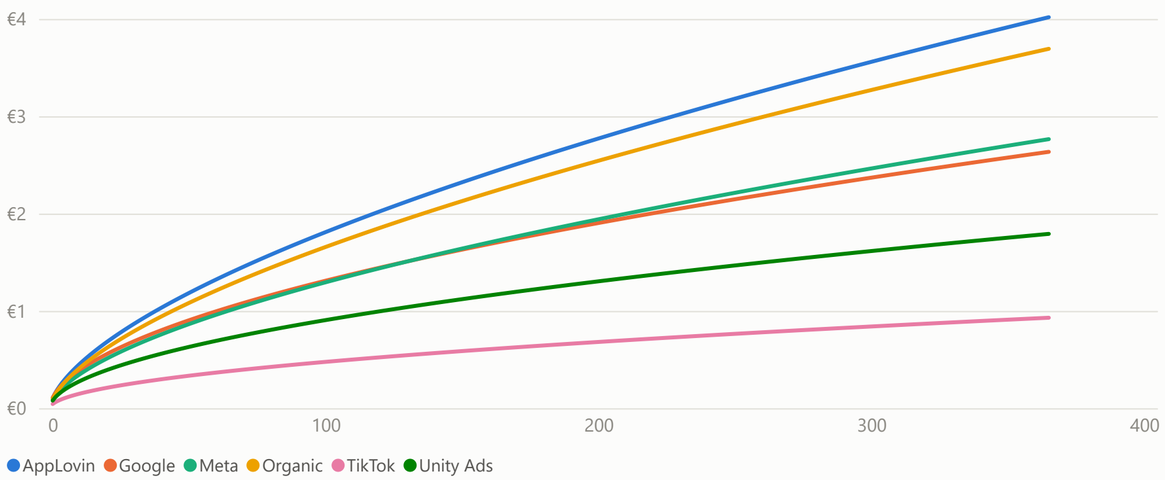


A 30-day fit predicts day-90 LTV within 6%: channels can be judged after one month

SIMULATED DATA

LTV(t) = a·t^b fitted on days 0-30 for each channel, checked against the observed day-90 value (holdout). The channel breaks even on the day its LTV curve crosses its CPI.

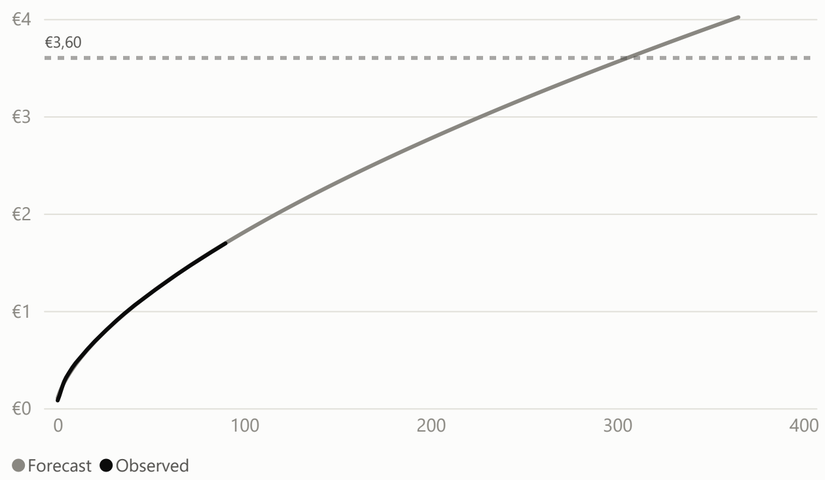
Forecast cumulative LTV per install by channel, EUR



Pick one channel

- (●) AppLovin
- ( ) Google
- ( ) Meta
- ( ) Organic
- ( ) TikTok
- ( ) Unity Ads

Selected channel: forecast (grey) vs observed (black) LTV, dashed = CPI



Forecast by paid channel (holdout error = fit on D0-30 vs actual D90)

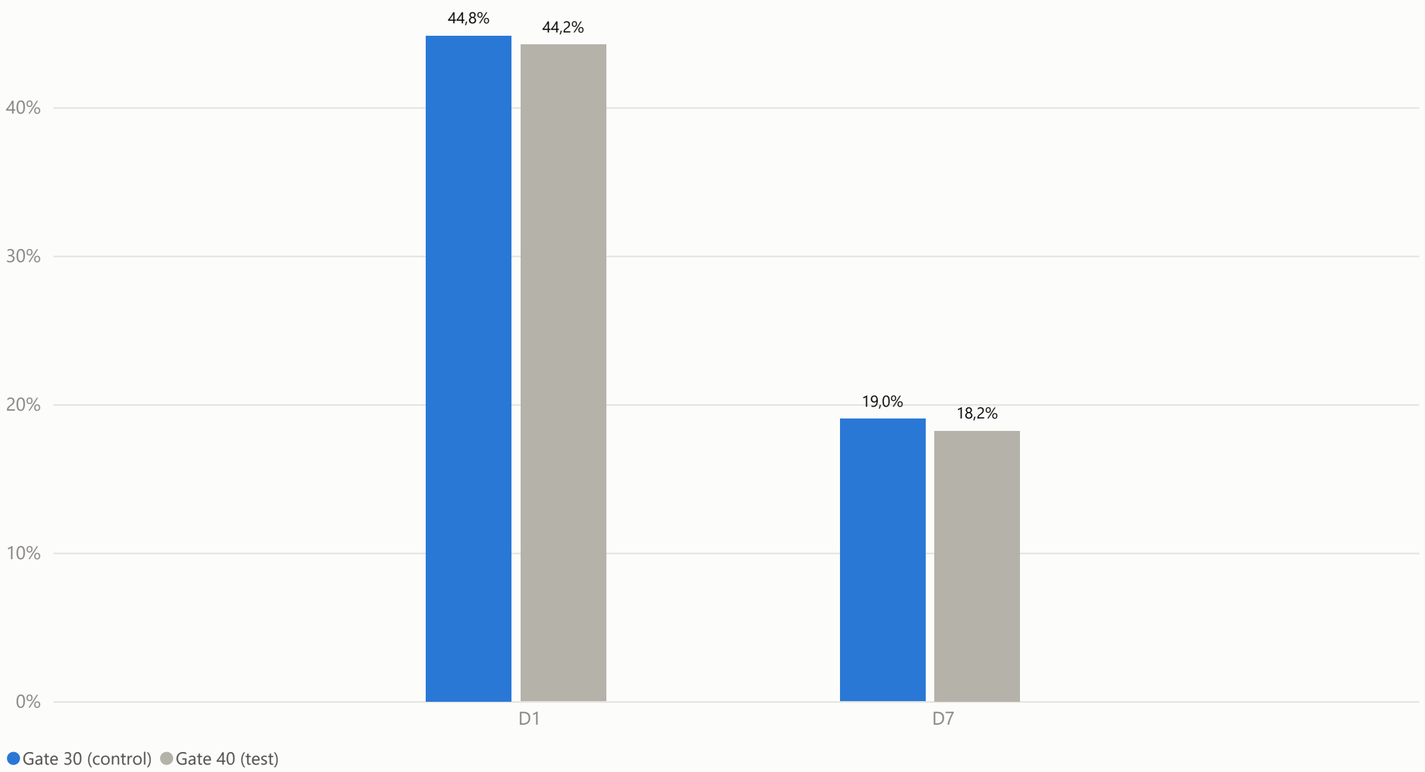
Channel	CPI (EUR)	LTV D90 actual	LTV D180 forecast	LTV D365 forecast	Holdout error D90
AppLovin	3,60	1,69	2,60	4,02	3,0%
Google	2,59	1,23	1,80	2,64	5,5%
Meta	3,00	1,22	1,83	2,77	5,0%
TikTok	1,70	0,45	0,65	0,93	-1,0%
Unity Ads	2,18	0,86	1,23	1,79	4,4%

Keep the first gate at level 30: moving it to 40 costs 0.8 pts of D7 retention

REAL DATA · COOKIE CATS

Two-proportion z-test on 90,189 randomised players. The D7 gap is significant (p = 0.002) and the bootstrap finds gate 30 ahead in 100% of resamples. The D1 gap is not significant (p = 0.07).

Retention rate by variant



D7 retention gap, gate 30 minus gate 40 (pts)

0,82 pts

D7 p-value (two-sided z-test)

0,0016

Test statistics (a CI that excludes 0 means the gap is significant)

Metric	Gate 30	Gate 40	Gap (pts)	95% CI low (pts)	95% CI high (pts)	p-value	P(gate 30 better), bootstrap
D1	44,8%	44,2%	0,59	-0,09	1,26	0,0744	96%
D7	19,0%	18,2%	0,82	0,31	1,33	0,0016	100%

Source: Cookie Cats mobile game A/B test, public dataset (90,189 players) · Independent Data Analyst case study · Not affiliated with Rovio